

Final Report

A Statistical Approach to Monitor Quantisation in Neural Network Training

09I03-03-V04-00562

Ing. Roman Budjač, PhD., 04/2026

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PROJECT IDENTIFICATION

Project Title: A Statistical Approach to Monitor Quantisation in Neural Network Training

Acronym: SAMQ-NN

Project Code: 09I03-03-V04-00312-00562

Program Name: Plán obnovy a odolnosti SR

Component: 9. Efektívnejšie riadenie a posilnenie financovania výskumu, vývoja a inovácií Plánu obnovy a odolnosti Slovenskej republiky

Investment: 3. Excelentná veda

Schema: State Aid Scheme to Support Research, Development, and Innovation under Component 9 of the Slovak Republic's Recovery and Resilience Plan No. SA.106633

Start of project implementation: 06/2024

Completion of the physical implementation: 05/2026

Principal Investigator: Ing. Roman Budjač, PhD.

ÚVOD

The project “A Statistical Approach to Quantization and Monitoring of Neural Network Training (SAMQ-NN)” addresses the issue of neural network model size, specifically the process of quantizing neural network models. The robustness of modern deep learning models complicates their deployment in segments with limited computational power, particularly in the IoT and resource-constrained devices. The quantization process reduces the memory and computational requirements of models, but it can lead to a loss of accuracy.

The project presents a concept for adapting neural network training for the quantization process based on statistical methods that compare the distribution of model weights before and after quantization. Using these methods, it is possible to monitor the training of the quantized model by tracking the similarity index of the weight distribution relative to the unquantized model at full precision. The training process modified in this way allows the user to monitor the progress and results of the training.

The goal of the project was to design a conceptual framework that would allow the quantization process of neural networks to be monitored using a similarity index.

PROJECT IMPLEMENTATION METHODOLOGY

The project methodology is based on comparing the weight distributions of quantized and non-quantized neural network models. Comparing distributions is a fundamental tool of statistical analysis that allows us to quantify differences between two data sets, track their evolution, and draw conclusions about their relationship. In the context of this project, this comparison is used to express the difference between the model's weight distribution before and after quantization, repeatedly throughout the training process.

Weights are a fundamental component of a neural network model. They represent the strength of the connections between neurons and determine the network's output for a given input. During training, the weights are adjusted using optimization algorithms, such as gradient descent and its variants, with the goal of minimizing the loss function, which is the difference between the expected and actual outputs. The distribution of weights in the network is related to its capacity and ability to capture relationships in the data. During quantization, weight values are converted from a higher precision to a lower one, thereby changing their distribution. Monitoring this change is therefore a suitable indicator of the impact of quantization on the model.

To compare weight distributions, three statistical measures were used in the project: Kullback-Leibler divergence, Jensen-Shannon divergence, and cosine similarity. These methods were first analyzed and tested on reference distributions and then applied to the weights of convolutional neural networks.

Comparing weight distributions has several uses within the project. It allows us to assess the difference in behavior between the quantized and unquantized models, since similar distributions generally correspond to similar accuracy, while significant differences may indicate a loss of accuracy. Furthermore, it allows us to assess the extent to which quantization has preserved the information contained in the weights of the original model—that is, the effectiveness of the quantization process itself. It also provides information on changes in the model's robustness, shifts in the relationship between bias and variance, and the suitability of the weight distribution for specific hardware.

The similarity index introduced by the project is based on statistical methods for comparing probability distributions. Comparing distributions is a fundamental tool of statistical analysis used across a wide range of fields, enabling the discovery of patterns, an understanding of the internal structure of data, and the drawing of conclusions about its nature. By comparing distributions, it is possible to quantify differences, monitor performance, optimize processes, and assess risks. There are several methods for comparing distributions, each with its own specific advantages and limitations, and their suitability depends on the specific situation. Within the project, these methods are used to calculate a similarity index between the weight

distributions of the quantized and unquantized models, thereby providing a quantitative measure of the degree of model degradation during training.

Based on these methods, the similarity index of the weight distribution relative to the non-quantized model is evaluated at full accuracy during the training of the quantized model. The index can be monitored in each epoch, providing ongoing information about the impact of decreasing bit depth on the model. The comparison of distributions itself is computationally inexpensive. The implementation was carried out in the TensorFlow/Keras framework, and the proposed method was validated on standard benchmark datasets.

Mathematical Formulation of the Similarity Index

The purpose of the index is to quantify how much the weight distribution of the quantized model deviates from the weight distribution of the original non-quantized model at full precision (FP32), and to monitor this deviation continuously during training. The starting point is the representation of the set of weights as a discrete probability distribution, to which statistical measures of the difference between two distributions are then applied.

Representing weights as a distribution. Let $\mathbf{w}^{\text{ref}}$ denotes the concatenated kernels of the convolutional and fully connected layers of the original model (biases and normalization parameters are not included), and let $\hat{\mathbf{w}}(t)$ denotes the effectively quantized model kernels at the end of epoch t . A common range of values is chosen for this pair of vectors, and it is divided into $K=100$ bins of equal width:

$$\ell = \min \left(\min_j w_j^{\text{ref}}, \min_j \hat{w}_j(t) \right), \quad u = \max \left(\max_j w_j^{\text{ref}}, \max_j \hat{w}_j(t) \right). \quad (1)$$

Before normalization, a small stabilization constant $\varepsilon = 10^{-10}$ is added to the weight counts in each bin to prevent zero probabilities and, consequently, the calculation of the logarithm of zero. The resulting distributions of the reference model P and the quantized model Q are

$$P_k(t) = \frac{h_k(\mathbf{w}^{\text{ref}}) + \varepsilon}{\sum_{j=1}^K (h_j(\mathbf{w}^{\text{ref}}) + \varepsilon)}, \quad Q_k(t) = \frac{h_k(\hat{\mathbf{w}}(t)) + \varepsilon}{\sum_{j=1}^K (h_j(\hat{\mathbf{w}}(t)) + \varepsilon)}, \quad (2)$$

where $h_k(\cdot)$ is the number of weights falling into the k th bin. Since the bin boundaries are determined by the common range of the pair, they are recalculated for each comparison, and both distributions therefore carry the epoch index t . Using the same boundaries for both P and Q ensures that the comparison is always consistent.

Measures of the difference between distributions. Three measures are used to compare P and Q , all based on the natural logarithm. Kullback–Leibler divergence

$$D_{\text{KL}}(P \parallel Q) = \sum_{k=1}^K P_k \log_2 \frac{P_k}{Q_k} \quad (3)$$

It is asymmetric and unbounded, and the magnitude of the value allows one to estimate the extent of the difference; it is therefore a sensitive indicator. The Jensen-Shannon divergence is its symmetric and bounded version. For a normal distribution with mean $M = 1/2 (P + Q)$, the following holds:

$$D_{\text{JS}}(P, Q) = 1/2 D_{\text{KL}}(P \parallel M) + 1/2 D_{\text{KL}}(Q \parallel M) \in [0, 1] \quad (4)$$

Because the base-2 logarithm takes values in the interval $[0, 1]$, it is symmetric and robust even when distributions overlap only slightly, making it a suitable measure for thresholding. Additionally, cosine similarity is calculated, which compares the shape (orientation) of distributions independently of their magnitude:

$$\cos(P, Q) = \frac{\sum_{k=1}^K P_k Q_k}{\sqrt{\sum_{k=1}^K P_k^2} \sqrt{\sum_{k=1}^K Q_k^2}} \quad (5)$$

Similarity Index.

The primary metric monitored during training is the Jensen-Shannon divergence between the reference and current quantized distributions, evaluated globally across all kernels: $S(t) =$

$$D_{JS}(P(t), Q(t)). \quad (6)$$

The same definition can also be applied to the kernels of the individual layers l to obtain a per-layer index $S_\ell(t) = D_{JS}(P_\ell(t), Q_\ell(t))$, which shows which layers are most affected by quantization.

Weight quantization.

Effectively Quantized Cores $\hat{w}$ are generated by uniform, symmetric quantization across the output channels. For channels c and b bits, the quantization step and the quantized value

$$\Delta_c = \frac{\max_j |w_{c,j}|}{2^{b-1}-1}, \quad \hat{w}_{c,j} = \Delta_c \cdot \text{clip}(\lfloor w_{c,j}/\Delta_c \rfloor, -(2^{b-1}-1), 2^{b-1}-1), \quad (7)$$

where $\lfloor \cdot \rfloor$ rounding to the nearest whole number and clipping to the permissible range. Each channel has its own step Δ_c , so the weights rest on the grid with a maximum of 2^b levels. In quantization-aware training, this operation is inserted into the forward pass, while a straight-through estimator (unit gradient) is used in the backward pass to ensure that the weights remain properly quantized throughout the training process.

Implementation of a similarity index in training management. The validity threshold of $\tau=0.01$ is based on the boundedness of the Jensen-Shannon divergence. The automatic bit depth selection scans through the candidates $B=\{4,6,8,16\}$ starting with the smallest and selects the first one for which the index does not exceed the threshold,

$$b^* = \min\{b \in B: D_{JS}(P, Q_b) \leq \tau\}, \quad (8)$$

and if none of them meet the threshold, it will retain the highest value. Index-controlled early termination stops the training when, during $S_{\text{pat}} = 3$ Over successive epochs, both the index and the accuracy stabilize, that is, when

$$|S(t) - S(t-1)| < \delta_s \quad \text{a súčasne} \quad a(t) - a(t-1) < \delta_a, \quad (9)$$

where $a(t)$ is the accuracy on the test set, $\delta_s = 10^{-3}$ and $\delta_a = 5 \cdot 10^{-3}$. The second condition is a sign condition: it is triggered when accuracy no longer increases noticeably, so that termination does not occur prematurely.

Histograms, divergences, and cosine similarity are calculated over concatenated kernels (bias and normalization layer parameters are omitted), with $K=100$ bins and a constant $\varepsilon=10^{-10}$. The reference is the original model in full precision (FP32), whose weights do not change during retraining; the index is evaluated at the end of each epoch within a callback and recorded in

the training history. The calculation itself consists of constructing a histogram and evaluating entropy or a scalar product, so its computational overhead is low compared to a single forward and backward pass through the network.

DESCRIPTION OF ACTIVITIES PERFORMED

The project was carried out from June 2024 to May 2026 and was divided into four work packages (WP1 through WP4), which were defined by four milestones (M1 through M4) and followed on from one another in sequence. Seventeen deliverables (D1 through D17) were produced during the project, and implementation proceeded in accordance with the project timeline.

As part of the first work package (WP1), an analysis and testing of suitable statistical methods for comparing the weight distributions of neural networks at various stages of quantization were conducted. Several methods for comparing probability distributions were evaluated, with a focus on the Kullback-Leibler divergence, the Jensen-Shannon divergence, and cosine similarity. Based on their properties, particularly the asymmetry and unboundedness of the Kullback-Leibler divergence compared to the symmetry and boundedness of the Jensen-Shannon divergence—the Jensen-Shannon divergence was selected as the most suitable for practical monitoring. The outputs of this phase were an introductory seminar (D1), an analytical report (D2), source code (D3), and a paper presented at a scientific conference (D4). Completion of this phase marked the achievement of the project's first milestone (M1). This analysis also showed that the statistical methods used did not impose a computational burden, and the comparison of distributions itself accounted for only a small fraction of the total computational effort expended during the neural network training process. The results were published in the paper “Layer-wise Statistical Analysis: A Time-Sensitive Framework for Implementing Statistical Methods in Neural Network Training Sessions” at the CSOC 2024 conference. This process is computationally inexpensive compared to the forward and backward passes of the network during training itself. The overhead of continuously evaluating the index therefore remains negligible and has no significant impact on the total training time.

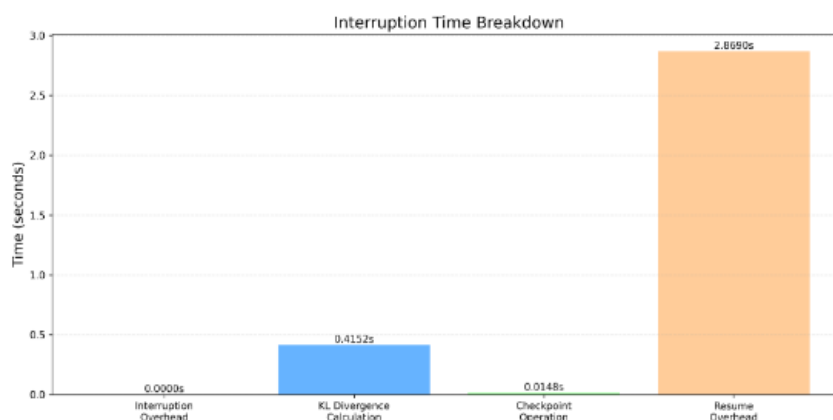
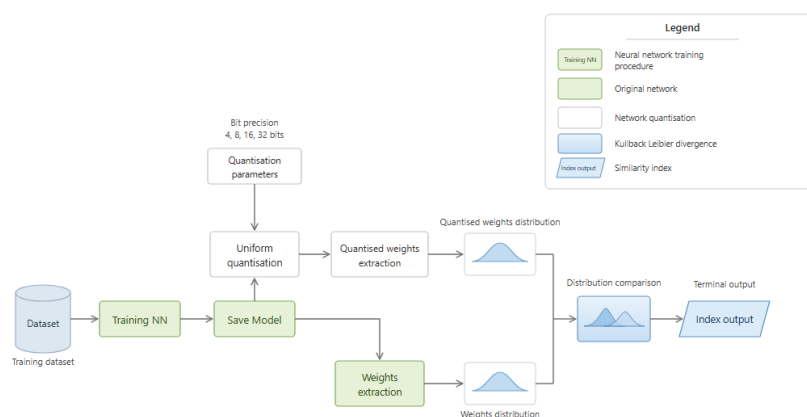


Fig. 1 Analysis of the Implementation of a Statistical Method in Neural Network Training.

Work Package 2 (WP2) focused on implementing the selected statistical method directly into the neural network quantization process. The Jensen-Shannon divergence was integrated into the comparison of the weight distributions of the quantized and unquantized models, enabling a quantitative analysis of changes in the weight distributions both during training and after quantization. Experiments were conducted on datasets to assess the impact of quantization on weight distributions and model performance. The outputs of this phase included a seminar (D5), a report (D6), source code made available on GitHub (D7), a conference paper (D8), and an interim project report (D9). This phase marked the achievement of the project's second milestone (M2). The implementation design diagram is shown in the figure below.



Obrázok 2 Návrh implementácie štatistických metód do procesu tréninga

The diagram below illustrates the operating principle of the proposed similarity index. The inputs are two models: a reference model with full precision (FP32) and a quantized model trained using a quantization method with a straight-through estimator, which ensures that its weights remain truly quantized throughout the entire training process. Distributions P and Q are constructed from the weights of both models, represented by histograms over a common range of values. The measure of the difference between these distributions is the Jensen-Shannon divergence, which constitutes the similarity index $S(t)$ itself and is evaluated at the end of each epoch via a callback. The index value is then used in two decisions: for the automatic selection of bit depth prior to training, where the lowest number of bits at which the index remains below a specified threshold is chosen; and for index-controlled early termination during training, which stops learning once the index and accuracy have stabilized. The index thus links analytical preparation with direct control of the training process.

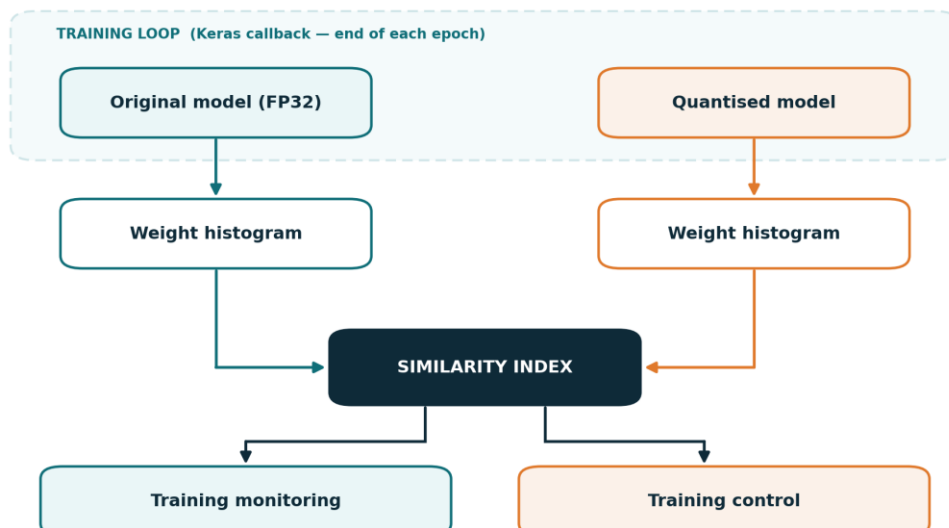


Fig. 3 How the similarity index implementation works.

The diagram below summarizes that the proposed similarity index is not a single, fixed quantity, but a tool that can be adjusted along four axes depending on its intended purpose at a given step in the solution process. The first axis is granularity: the index can be calculated globally—that is, across all model weights at once, yielding a single summary metric—or layer by layer, thereby revealing the most sensitive layer and providing a finer-grained signal suitable for mixed-precision strategies. The second axis is the role during training: in predictor mode, the index is used before training to select an appropriate bit depth; in monitor mode, it is evaluated continuously during quantization-aware training (QAT), at the end of each epoch. The third axis is the choice of reference: the current weight distribution can be compared either against a frozen reference model at full precision (FP32), thereby measuring the absolute deviation caused by quantization, or between consecutive epochs, thereby tracking epoch-to-epoch changes and allowing for the detection of convergence. The fourth axis is the decision-making method: a fixed threshold is used to select the bit depth, at which the index must remain below the value $JS \leq \tau$, and this mechanism can be supplemented by early training termination, which stops learning once the index has stabilized. These variants are not mutually exclusive but complementary: it is precisely through their combination that the method gains the ability to serve both as a predictor and as a monitor, covering the entire training process from parameter selection to controlled termination.

The third work package focused on the design and development of a new training strategy. Statistical comparison of weight distributions was integrated directly into the learning process in the form of a similarity index, based on which a proprietary strategy called Statistically-Monitored Training (SMT) was proposed. This strategy includes automatic

selection of the bit depth based on the index threshold and the option to terminate training early once the weight distribution relevant for quantization has stabilized.

How to use the index - variants

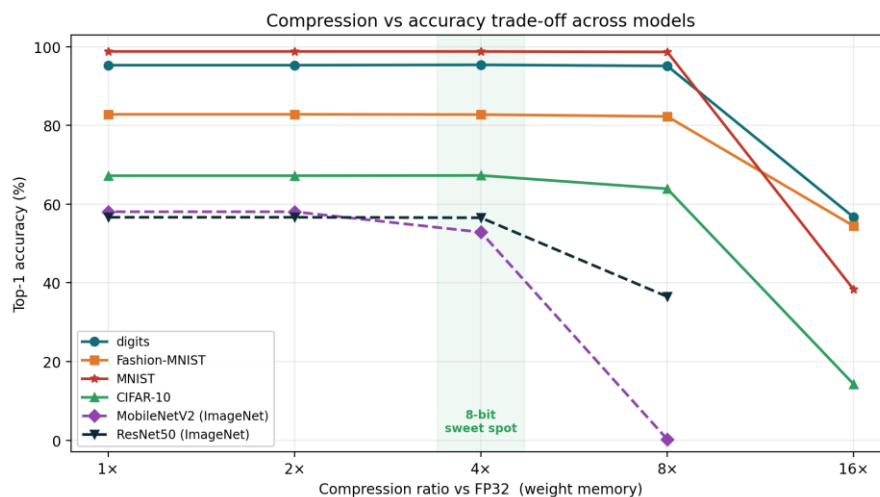
	Granularity	global (whole model)	vs	per-layer (most sensitive layer)
	Role in training	predictor (pre-training — bit-width selection)	vs	monitor (during QAT — per epoch)
	Reference	frozen FP32 reference	vs	epoch-over-epoch (inter-epoch change)
	Decision-making	fixed threshold $JS \leq \tau$ (bit-width selection)	vs	+ early stopping at stabilisation

Fig. 4 Applications of the Similarity Index

Work Package 4 (WP4) involved a comprehensive verification of the proposed concept using standard datasets and the final validation of the implemented methods. The similarity index showed a monotonic decrease with decreasing bit depth, and the proposed SMT strategy achieved accuracy comparable to the reference 8-bit approach while offering a higher

compression ratio. At the same time, the limits of the proposed index's applicability at extremely low bit depths were clearly defined.

*** Translated with www.DeepL.com/Translator (free version) ***



Obrázok 5 Kompresia vs. presnosť

For the purposes of experimental validation, a dedicated device was built as part of the project to test quantization and monitor it in real time. The device is based on hardware suitable for both the development and deployment of deep learning and artificial intelligence models, specifically on NVIDIA Jetson Orin Nano Super platforms. The goal of this validation was to assess the usability of the proposed method under conditions that differ from standard computing infrastructure and to test it directly on target devices with limited computing power. However, it should be emphasized that this project was a proof of concept, the purpose of which was to verify the feasibility and basic functionality of the approach on such hardware. Optimization for specific deployments, as well as

more extensive performance testing, go beyond the scope of the project and represent a natural area for follow-up work.

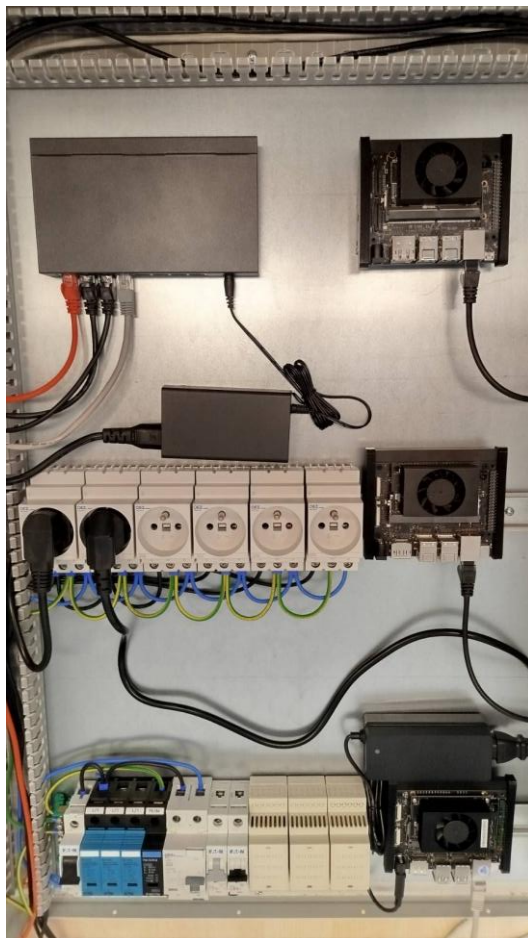


Fig. 6 Verification of the project methodology on hardware.

The practical significance of the results achieved lies in the potential for deploying quantized neural networks on devices with limited computational and memory resources, such as microcontrollers and embedded systems. Reducing the bit depth of the weights leads to a significant reduction in model size and memory requirements, which are key prerequisites for running models on such hardware.

PROBLEMS, RISKS, AND MEASURES TAKEN:

Risk 1: Insufficiently powerful hardware for model training (WP1 through WP4), medium probability, medium severity.

Mitigation 1: Use of the host institution's prepaid cloud computing services (LightningAI). Development took place in a CPU environment, and training runs were validated on GPU instances with regular saving of checkpoints (local PC).

Risk 2: Statistical methods may not be suitable for comparing weight distributions (WP1), medium probability, medium severity.

Mitigation 2: The suitability of statistical methods was verified on reference distributions with known expected outcomes before applying them to model weights on a small dataset.

Risk 3: Statistical methods may not yield the expected results in the case of quantization (WP2 and WP3); medium probability, medium severity.

Mitigation 3: Iterative testing and gradual refinement of the methodology; comparison of the effectiveness of multiple methods at different bit depths.

Risk 4: Statistical methods may significantly slow down training (WP4); low probability, medium severity.

Mitigation Measure 4: The computational overhead associated with comparing distributions is generally low and negligible relative to the computational complexity of the training itself.

Risk 5: Inadequate achievement of planned publication outputs; medium probability, medium severity.

Mitigation Measure 5: Publication outputs were planned on an appropriate scale, taking into account the project duration and the researcher's career stage.

No major issues that would have negatively affected the project's progress were identified during its implementation. The risks listed in this section are based on the risk analysis conducted during the project planning phase and documented in the project proposal. These risks were continuously monitored throughout the project's implementation, and appropriate measures were proposed in advance for each of them and implemented as necessary.

PROJECT IMPLEMENTATION PROCESS

The project is divided into four work packages (WP1 through WP4), which are defined by four milestones (M1 through M4), and includes deliverables D1 through D17. The first milestone (M1) involves the analysis and testing of suitable statistical methods for comparing weight distributions. The second milestone (M2) involves analyzing the implementation of the weight distribution comparison. The third milestone (M3) ensures the implementation of a neural network training strategy based on a statistical method. The fourth milestone (M4) covers the evaluation of the concept of statistical methods for the neural network training process.

During the project, all objectives of Work Packages WP1 through WP4 were successfully achieved. An analysis and testing of suitable statistical methods for comparing the weight distributions of neural networks were conducted, and the Kullback-Leibler divergence, Jensen-Shannon divergence, and cosine similarity were selected as the most appropriate methods. Subsequently, JS was implemented into the quantization process, enabling a detailed analysis of changes in the weight distribution both during training and after quantization. Based on these methods, a new training strategy (Statistically-Monitored Training, SMT) was proposed and validated; this strategy integrates statistical comparison of weight distributions into the learning process in the form of a similarity index, thereby enabling the monitoring of the impact of decreasing bit depth on the model. The chosen methodological approach proved to be suitable and, at the same time, clearly defined the limits of the proposed index's applicability at extremely low bit depths. This established an experimental and methodological foundation for the further development of statistically monitored training of quantized neural networks.

Project outputs as specified in the "Project Description" appendix:

WP1

D1: Seminar. Seminar on research topics and the presentation of research ideas. The report will be available on GitHub.

D2: Report. Analysis of appropriate statistical methods. A written summary will be available on GitHub.

D3: Source code. Testing of statistical methods for comparing distributions. The code in the Jupyter Notebook environment will be available on GitHub.

D4: Publication at a scientific conference. Conference proceedings. Participation in an international scientific conference whose proceedings are indexed in the Scopus database.

WP2

D5: Seminar. A seminar dedicated to the research task (WP2) and the presentation of research ideas. The report will be available on GitHub.

D6: Report. Analysis of the application of statistical methods in the quantization process. A written summary will be available on GitHub.

c

D7: Source code. Implementation of statistical methods in the quantization process. The code in the Jupyter Notebook environment will be available on GitHub.

D8: Publication at a scientific conference. Publication in the conference proceedings. Participation in an international scientific conference whose proceedings are indexed in the Scopus database.

D9. Interim report..

WP3

D10: Source code. Analysis comparing the distribution of implementation weights in the network training process. A written summary will be available on GitHub.

D11: Seminar report. Seminar dedicated to the research task and the presentation of research ideas. The code in Jupyter Notebook will be available on GitHub.

D12: Seminar report. Implementation of a new neural network training strategy based on a statistical method. The code in Jupyter Notebook will be available on GitHub.

D13: Report. The validated concept and its code (Jupyter Notebook) will be available on GitHub.

D14: Seminar report. Presentation of project results at the seminar. The presentation will be available on GitHub.

D15: Seminar, report. Implementation of a new neural network training strategy based on a statistical method. A written summary will be available on GitHub.

D15: Finalization of the proposed method and preparation of a research summary in the proposed area. A written summary will be available on GitHub.

D16: Publication in a scientific journal. Publication in a journal ranked at least in the Q2 quartile according to the Web of Science (WOS) classification.

D17: Final report

Assessment of the Achievement of Project Outputs:

During the project's implementation, all project deliverables (D1 through D17) were successfully completed. Seminars were held to present the interim results of the project, reports, summaries, and other documents were prepared, and the project's source code was published, including its release on the GitHub repository. The interim status of the project was summarized in the interim project report (D9).

As part of the dissemination activities, the project's minimum requirements for key outputs were met, namely two conference papers indexed in the SCOPUS database (D4, D8) and an article in a scientific journal ranked in the Q2 quartile according to the Web of Science (D16).

This fulfilled the project's minimum requirements regarding publication. The implementation of statistical methods, their integration into the quantization process, and the proposed training strategy were documented in the form of source code and accompanying reports and summaries. The final output of the project is this final report (D17), which summarizes the methodology, activities carried out, results achieved, and the fulfillment of the project's objectives, outputs, and milestones.

Project Milestones:

M1 - Analysis and testing of suitable statistical methods for comparing weight distributions.

M2 - Analysis of the comparison of weight distributions upon implementation into the network training process

M3 - Implementation of a new neural network training strategy based on a statistical method

M4 - Verification of the concept through the implementation of statistical methods into the training process

Assessment of Project Milestone Achievement: All four project milestones (M1 through M4) were fully achieved, confirming the smooth progress of the project in accordance with the work package schedule.

PROJECT PUBLICITY

The project results were disseminated to the scientific and professional community through publications in peer-reviewed and impact journals, presentations at international scientific conferences, and publicly available outputs made accessible in the project repository. The publications adhered to open access principles. The project outputs are available on the GitHub repository: <https://github.com/romanbudjac>, under the project name: <https://github.com/romanbudjac/samqnn>.

Making the project's outputs available is part of the open science concept, which aims to make scientific knowledge, data, and software tools accessible to the widest possible audience, thereby increasing the transparency, verifiability, and reusability of research results. Open access to publications, open source code, and publicly available outputs enable their reproducibility and further development by other researchers. Open science is also consistent with the principles of research funding at the European Union level.

Overall, it can be concluded that the project's publications and scientific outputs exceed the scope of the minimum planned outputs. The project has produced scientific articles, conference papers, seminars, and source codes, thereby demonstrating its contribution.

Publications related to the project:

List of Publications and Research Outputs

a) publications indexed in Web of Science

- BUDJAČ, Roman – VALÁŠEK, Maroš – BELÁNY, Pavol – GAŠPAR, Gabriel.
Multiservice-Oriented Framework for Enhanced Data Collection and
Visualization in Smart City Applications. In *IEEE Access* [online]. 2026, roč. 14, s.
5133–5151 [cit. 2026-06-21]. ISSN 2169-3536. Dostupné na:
<https://doi.org/10.1109/ACCESS.2026.3650820>.
- Gašpar G.; Budjač R.; Valášek M.; Bartoň M.; Meravý T.; Strémy M.: Digitizing
SMEs in the EU: A Scalable Model for Retrofitting Machinery to Industry 4.0.
Applications in Engineering Science, roč. 23, art. č. 100230, 2025. DOI:
<https://doi.org/10.1016/j.apples.2025.100230> Výstup spĺňa požiadavku na
článok kategórie v databáze WOS Q2.

- GAŠPAR, Gabriel – BUDJAČ, Roman – ŠEDIVÝ, Štefan – STRÉMY, Maximilián – BENKA, Denis – NEMLAHA, Eduard – ELIÁŠ, Rastislav. Innovative Grouting Process Monitoring: Design and Implementation of a Measurement Device With Datalogger Function. In *IEEE Access* [online]. 2025, roč. 13, s. 19340–19352 [cit. 2026-06-21]. ISSN 2169-3536. Dostupné na: <https://doi.org/10.1109/ACCESS.2025.3534816>.
- VALÁŠEK, Maroš – BUDJAČ, Roman – HANZELY, Marek – VRČEK, Neven. Cloud-Based Modular System for Acquisition and Visualization of OCO-2 Remotely Sensed CO2 Data. In *Communications – Scientific Letters of the University of Žilina* [online]. 2026, roč. 28, č. 1, s. C10–C21 [cit. 2026-06-21]. ISSN 2585-7878. Dostupné na: <https://doi.org/10.26552/com.C.2026.013>

b) Publications from international scientific conferences indexed in SCOPUS

Budjač R.; Valášek M.: Layer-Wise Statistical Analysis: A Time-Sensitive Framework for Implementing Statistical Methods in Neural Network Training Sessions. Lecture Notes in Networks and Systems, roč. 1558 LNNS, s. 428–445 2026. DOI: https://doi.org/10.1007/978-3-032-00236-5_32 - . *The paper meets the requirements for a conference paper in a proceedings volume indexed in the SCOPUS database.*

BENKA, Denis – GASPAR, Gabriel – BUDJAC, Roman – ELIAS, Rastislav – SKOVAJSA, Martin. Design and Implementation of the Grouting Process Measurement Device with a Datalogger. In *2024 21st International Conference on Mechatronics – Mechatronika (ME)* [online]. [Piscataway] : IEEE, 2024, s. 1–6 [cit. 2026-06-21]. Dostupné na: <https://doi.org/10.1109/ME61309.2024.10789730> *The paper meets the requirements for a conference paper in a proceedings volume indexed in the SCOPUS database.*

BENKA, Denis – GASPAR, Gabriel – NEMLAHOVA, Veronika – NEMLAHA, Eduard – BUDJAC, Roman. Proposal of a Remote-Control System for Building Lighting Management. In *2024 21st International Conference on Mechatronics – Mechatronika (ME)* [online]. [Piscataway] : IEEE, 2024, s. 1–5 [cit. 2026-06-21]. Dostupné na: <https://doi.org/10.1109/ME61309.2024.10789644>.

The paper meets the requirements for a conference paper in a proceedings volume indexed in the SCOPUS database.

BUDJÁČ, Roman – VALÁŠEK, Maroš – BELÁNY, Pavol – HRABOVSKÝ, Peter. Runtime

Monitoring and Correction of INT8 Quantized Models Under Distribution Shift. In

SILHAVY, Radek (ed.). Cham : Springer, 2026, Lecture Notes in Networks and Systems.

ISSN 2367-3370. ISBN, Letter of acceptance.

The paper meets the requirements for a conference paper in a proceedings volume indexed in the SCOPUS database.

Other conference proceedings indexed in SCOPUS - *currently in print*:

- BUDJÁČ, Roman – GAŠPAR, Gabriel – VALÁŠEK, Maroš – BARTOŇ, Martin.
Proposal for a weather monitoring system utilizing IoT technologies. In
*Proceedings of the 35th Central European Conference on Information and
Intelligent Systems (CECIIS 2024), Varaždin, Croatia, September 18–20, 2024*
[online]. Varaždin : University of Zagreb, Faculty of Organization and Informatics,
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PROJECT RESULTS AND OUTPUTS

The implementation of the project successfully achieved its main objective, which was the statistical monitoring of the quantization process during the training of neural networks by comparing the weight distributions of the quantized and unquantized models. The project also made it possible to comprehensively integrate the analysis and testing of statistical methods, their implementation into the quantization process, the design of a new training strategy, and the experimental validation of the proposed concept on standard datasets.

In the first phase of the project, an analysis and testing of suitable statistical methods designed to compare the weight distributions of neural networks at various stages of quantization were conducted. Several methods for comparing distributions were evaluated, and the Kullback-Leibler divergence, Jensen-Shannon divergence, and cosine similarity were selected as the most suitable. Testing was conducted on several datasets (sklearn digits, MNIST, Fashion-MNIST), which allowed for an assessment of how each method behaved at different quantization levels.

Based on the results of the analysis, the Jensen-Shannon divergence was subsequently implemented directly into the neural network quantization process. This implementation enabled a detailed analysis of changes in the weight distribution during training and after quantization, and laid the foundation for a quantitative measure of the difference between the quantized and unquantized models.

Another significant result was the design and validation of a new training strategy (Statistically-Monitored Training, SMT), which integrates statistical comparison of weight distributions into the learning process in the form of a similarity index. Based on the evolution of the index during training, an early stopping strategy was proposed, which was typically activated around the seventh to eighth epoch and allowed for a reduction in training time without a significant loss of accuracy. The computational overhead of continuously evaluating the index was negligible (on the order of 0.24% per epoch), making the proposed approach computationally efficient.

The final phase of the project involved a comprehensive validation of the proposed concept on standard datasets (MNIST, Fashion-MNIST, CIFAR-10, ImageNET). At low bit depths, the quantized models trained using the quantization-aware training (QAT) method maintained accuracy comparable to that of the unquantized model at full precision (FP32), which achieved approximately 96% to 97%. The functionality of the implemented solution was also verified by actually exporting the model to the int8 format (TFLite). The similarity index exhibited consistent monotonic behavior, with the Jensen-Shannon divergence increasing as the bit depth decreased. At the same time, the limits of the proposed index's applicability were clearly defined at extremely low bit depths, where, in the event of model collapse, the index may not reliably indicate a loss of quality.

The project yielded new insights into the monitoring of neural network quantization using statistical methods. From a technical standpoint, a similarity index based on Jensen-Shannon divergence was developed and validated, along with a new training strategy called SMT, which is computationally efficient and can be applied directly in the training process of quantized neural networks. From a scientific perspective, the project contributed to a better understanding of the relationship between decreasing bit depth and the degree of divergence in the weight distributions of quantized and unquantized models, with a particular contribution being the definition of the limits of applicability of the proposed index. From an applied perspective, the project's results can serve as a basis for the continuous monitoring and control of the quantization process, including early termination of training and optimization of bit depth when deploying neural networks on devices with limited computational resources.

FINAL PROJECT EVALUATION

From a comprehensive perspective on the entire course of the project, it can be concluded that the project progressed systematically in accordance with the structure of the work packages (WP1 through WP4), each of which had defined sub-objectives, milestones, and planned deliverables. The individual phases followed on logically from one another and gradually covered the entire scope of the project, from the initial analytical preparation of the quantization process and the possibilities for using statistical methods in the quantization process (WP1), through the design of the proposed concept (WP2) and its integration into the training process (WP3), to the final validation of the proposed concept (WP4). Each subsequent step built upon the results of the previous one and contributed to the achievement of the project's overall objective.

During implementation, all defined project objectives were gradually achieved, and all milestones were met at the respective checkpoints. The fulfillment of objectives and milestones is thus not tied to a single phase but represents a coherent and self-contained whole documented by a complete set of project deliverables (D1 through D17). This consistency confirms that the project was fully completed, with all objectives and milestones. The project objectives and all sub-tasks arising from the work packages were successfully completed.

The individual specific objectives of the project were also fulfilled. In the introductory section, the quantization technique itself was analyzed; subsequently, the possibilities for using statistical methods were examined, and methods suitable for comparing the weight distributions of the quantized and non-quantized models were identified. On this basis, a conceptual framework was defined for the comparative analysis of unquantized and quantized neural network models using statistical methods. A statistical approach to comparing weight distributions suitable for deep learning in the context of the quantization process was proposed. This concept was ultimately validated on standard datasets. All substantive objectives of the project were thus fulfilled, and the project can therefore be considered successfully completed.

The proposed solution also involves several aspects that further define its scope and, at the same time, indicate room for further development beyond the scope of the project. The presented concept (framework) is based on an indicator (similarity index) that monitors the quantization process using statistical methods applied to the model weights; therefore, it most reliably captures precisely those changes that manifest themselves in the weights.

Linking the similarity index directly to model accuracy and monitoring them together has proven to be a suitable and validated approach during implementation, as their informative value complements each other. The index thus naturally serves as a supplement to standard quality metrics and broadens the perspective on model behavior. Since accuracy is one of the indicators of model degradation, the method also offers parameters that allow it to be

adapted to specific conditions, making it possible to flexibly tune it for various types of models and tasks. Validation to date on a representative set of models and datasets has confirmed the effectiveness of the proposed approach and, at the same time, opens up natural opportunities for its further expansion. These findings represent a natural delimitation of the current scope of the solution and set the direction for follow-up work.

In terms of content, the project has resulted in a proposed approach—a similarity index—for monitoring the quality of neural network quantization (concept, framework), whose main benefits include low computational complexity and the ability to be directly integrated into the solution process without a drastic impact on its performance (<1%), as well as the detection of significant changes in model weights (monitoring of the quantization process).